Day3 演習4

固定点

```
import sympy as sp

# 変数と定数
x = sp.Symbol('x')
y = sp.Symbol('y')

a = 0.06
b = 0.6

# 交点の座標を求める
solutions = sp.solve( [-x + a * y + (x ** 2) * y, b - a * y - (x ** 2) * y]  , ( x ,
                     y ) )

print(solutions)
```

実行結果

```
[(0.600000000000000, 1.42857142857143)]
```

線形化・ヤコビ行列・固有値

```
import sympy as sp

x, y, a, b = sp.symbols("x y a b", real=True, positive=True)

a_val = 0.06
b_val = 0.6

f = -x + a * y + x**2 * y
g = b - a * y - (x ** 2) * y

# ヌルクラインと固定点
solutions = sp.solve( [f, g]  , ( x , y ) )
fix_x , fix_y = solutions[0]

# ヤコビ行列
J_sym = sp.Matrix( [ [ f.diff( x ) , f.diff( y ) ] , [ g.diff( x ) , g.diff( y ) ] ] )
J_num = J_sym.subs( [ ( x , fix_x ) , ( y , fix_y ) , ( a , a_val ), ( b , b_val ) ] )

print( J_sym )
print( J_num )
print( J_num.eigenvals() )
```

実行結果

```
Matrix([[2*x*y - 1, a + x**2], [-2*x*y, -a - x**2]])
Matrix([[0.714285714285714, 0.420000000000000], [-1.71428571428571,
        -0.420000000000000]])
{0.147142857142857 - 0.631148936140937*I: 1, 0.147142857142857 + 0.631148936140937*I: 1}
```